

UConn

School of Business

Lab 1 — First Commit

Tonight in 20 steps

OPIM 5512 · Module 1 · Data Science using Python

The deliverable tonight is a REPO, not a notebook.

The deal

You and a partner each analyze a DIFFERENT half of the same problem — one takes the weather, one takes the electricity demand.

Neither of you can answer the question alone. The only way to the analysis is to merge your work together.

Then you write it up for a manager who doesn't know what a p-value is.

That's the lab. No scraping, no modeling tonight — those are Modules 2 & 3. Week 1 is the workflow you'll use all semester.

 **No AI tonight — no Claude, ChatGPT, Copilot, or Colab autocomplete. Type it yourself.**

Git is muscle memory, and you're about to approve real code. Module 3 is the whole AI unit. Not never — not yet.

Stack check

GitHub account + GitHub Desktop installed and signed in

Google Colab opens in your browser

Your campus station — Hartford = KBDL, Stamford = KBDR

Have the instructions open beside Colab — every click is in there, in order

Pick roles now — Partner A = weather, Partner B = demand

Odd one out pairs with Dave. Online / solo? Two accounts, or pair over Teams.

Make the shared repo

- 1 · Pair up — decide who's Partner A (weather) / B (demand)
- 2 · Partner A: create a new PUBLIC repo for the pair
- 3 · Partner A: add Partner B as a collaborator → B accepts the invite
- 4 · Partner A: turn on branch protection for main (PR + 1 approval)
- 5 · Both: clone the repo ONCE in GitHub Desktop
- 6 · Each: make your branch (dev-weather / dev-demand) → Publish it

Clone once, reuse forever. Branch protection = the two-person gate — you can't approve your own PR.

Clean your half, open one PR

- 7 · Open your starter in Colab (new tab) → run cell 1, LOOK at the raw data
- 8 · Together: write the data dictionary (contract) in README — names, units, timestamp
- 9 · Clean it → File → Save to your branch (drop a #TEST to prove it saved)
- 10 · Save your CSV and your plot; download both
- 11 · GitHub Desktop → Show in Explorer → drag CSV→data/clean/, plot→images/
- 12 · Commit (real message) → Push
- 13 · Open ONE pull request: notebook + contract + CSV + plot

Restart & run all → Clear all outputs before it goes in. Same path every save, or you make a duplicate.

The two-way loop

14 · Request your PARTNER as reviewer

15 · Review your partner's PR — read the diff, then Approve

16 · Merge both PRs → delete the branches

↳ whoever merges the README SECOND resolves the merge conflict — keep both halves

17 · Both: switch to main → Fetch → Pull — now you each have BOTH files

Resolve the conflict FIRST, then approve. Clear all three markers: <<<<<< ===== >>>>>>

Answer the manager

18 · Together (one screen): open the joint EDA notebook → run the JOIN → find 3 things (one surprising)

19 · Write REPORT.md (from the template) + save figures → branch → PR → partner reviews → merge

20 · Read-out: one figure on screen, one sentence — check the network graph shows the loop both ways

A number the manager gets: “each degree above 75°F adds roughly ___ MW” beats “ $r = 0.57$.”

Don't panic

Commit early. Once something's committed, it's essentially impossible to lose.

Fetch & Pull can't hurt you. Fetch = peek, Pull = download. Neither overwrites your work.

The scary buttons are the Discard ones — and you won't be clicking those.

main rejecting your push is the protection working — you're on main when you should be on your branch.

Deleted a branch by mistake? A merged PR has a Restore-branch button; git reflog recovers the rest.

Online / solo students

Pair over Teams if you can — each on your own account, one owns the repo and adds the other. Real review, real contract.

Otherwise, two accounts (needs a second email) — play both A and B yourself; you'll hit every step, including approving as the other account and resolving the conflict.

Simplest solo — one account with Required approvals = 0: branch → PR → merge yourself (skips the approval gate).

Either way — join the live session on Teams, and Friday office hours are highly recommended.

Definition of done

- ☐ Both partners are collaborators; branch protection on main
- ☐ README.md with a data dictionary for BOTH files
- ☐ data/clean/weather_hourly.csv AND data/clean/demand_hourly.csv
- ☐ Both cleaning notebooks + the joint EDA notebook
- ☐ REPORT.md — three findings (one surprising), figures, honest caveats, plain English
- ☐ ≥ 3 merged pull requests, branches deleted, both of you authoring AND reviewing
- ☐ A network graph showing the loop going both ways